

CHICAGO HARRISON-DEVER CRIB, IL, USA

WMO: 997338

Lat: 41.916N

Lon: 87.572W

Elev: 663

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.1	(c) 7.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 42.2	(k) 23.8	(l) 39.8	(m) 26.4	(n) 19.7	(o) 300	(p) 0.747

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.2	(c) 87.1	(d) N/A	(e) 84.2	(f) N/A	(g) 81.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 18.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 36.9	(b) 33.5	(c) 30.0	(e) -3.2	(f) 90.5	(g) 7.5	(h) 3.9	(i) -8.6	(j) 93.2	(k) -13.1	(l) 95.5	(m) -17.3	(n) 97.7	(o) -22.8	(p) 100.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	24.9	26.2	37.1	46.7	57.1	67.0	73.0	72.5	66.3	54.5	42.7	31.2
	DBStd	18.87	10.54	10.14	9.96	7.87	8.68	7.59	5.93	4.47	6.49	7.96	9.64	9.25
	HDD50	2925	777	667	418	157	27	0	0	0	0	42	253	584
	HDD65	6214	1242	1087	867	551	276	68	8	4	57	339	668	1047
	CDD50	2952	0	1	17	59	248	511	712	696	490	181	35	2
	CDD65	765	0	0	1	2	32	128	256	235	98	13	0	0
	CDH74	3793	0	0	2	14	197	712	1479	930	409	50	0	0
	CDH80	885	0	0	0	0	32	150	425	178	95	5	0	0
Wind	WSAvg	15.8	18.2	17.2	16.4	16.9	15.1	13.6	12.5	12.4	14.5	17.4	18.6	17.2
Precipitation	PrecAvg	35.7	1.7	1.5	2.5	3.6	3.4	3.6	3.7	4.4	3.5	2.6	2.8	2.3
	PrecMax	49.3	4.4	4.4	5.9	7.7	7.3	10.0	8.3	17.1	14.2	8.5	8.2	8.5
	PrecMin	22.3	0.1	0.2	0.6	0.7	0.3	0.9	1.1	0.7	0.0	0.2	0.5	0.3
	PrecStd	6.4	1.1	0.9	1.2	1.7	1.8	2.0	1.7	3.3	2.7	1.8	1.8	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.1	72.0	76.4	84.4	89.1	93.5	88.1	87.3	79.4	67.5	57.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	48.8	50.1	66.0	70.7	79.6	84.0	88.3	85.0	82.4	74.1	62.9	51.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	43.6	43.4	59.3	66.0	75.7	80.9	84.9	81.6	78.4	69.9	59.1	47.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	39.0	38.2	51.3	60.2	71.3	78.1	81.6	78.7	75.2	65.9	55.4	43.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.5	10.8	11.6	12.7	11.3	9.2	7.9	9.0	9.9	10.3	9.5
		MCDBR	14.8	17.1	22.2	23.1	19.1	16.1	14.7	12.6	13.8	14.0	14.2	13.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.303	0.322	0.348	0.403	0.432	0.440	0.448	0.445	0.416	0.361	0.321	0.304
	taud		2.461	2.388	2.381	2.262	2.238	2.267	2.243	2.251	2.296	2.450	2.515	2.504
	Ebn at Noon		269	280	285	275	269	266	262	259	259	263	260	259
	Edh at Noon		24	30	34	41	43	42	43	41	37	28	23	21
All-Sky Solar Radiation	RadAvg		490	760	1129	1461	1667	1917	1937	1681	1382	908	571	410
	RadStd		55	98	85	120	140	124	90	99	117	97	60	46

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page